https://xtremepape.rs/

Question	Answer	Marks
1	**Defining the problem**	
	L is the independent variable and I is the dependent variable **or** vary *L* and measure I	**1**
	keep *E* constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • circuit diagram with power supply connected to ends C • ammeter in series with power supply and conductors • correct symbol for ammeter and power supply	**1**
	circuit diagram with voltmeter correctly positioned to measure *E* across the power supply	**1**
	use a rule(r) to measure *L* **and** *x*	**1**
	use a micrometer/calipers to measure *y*	**1**

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Question	Answer	Marks
1	**Method of analysis**	
	plot a graph of $\frac{1}{I}$ against L or equivalent (e.g. L against $\frac{1}{I}$) (Do not accept log graphs.)	**1**
	$P = \frac{\text{gradient} \times AE}{2}$ (for L against $\frac{1}{I}$: $P = \frac{AE}{2 \times \text{gradient}}$)	**1**
	$Q = \frac{y\text{-intercept} \times Ey^2}{x}$ (for L against $\frac{1}{I}$: $Q = -\frac{y\text{-intercept} \times 2Py^2}{Ax}$ **or** $Q = -\frac{y\text{-intercept} \times Ey^2}{\text{gradient} \times x}$)	**1**

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